

Rotational fission of contact binary asteroids

D.J. Scheeres

Department of Aerospace Engineering, University of Michigan, 1320 Beal Ave, Ann Arbor, MI 48109-2140, USA

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Abstract

The energetics and dynamics of contact binary asteroids as they approach and pass the rotational fission limit is studied. We presume that the asteroids are subject to an external torque, such as from the YORP effect, that increases their angular momentum. Furthermore, we assume the asteroids can be described by a fairly minimal model comprised of a sphere and ellipsoid resting on each other. The minimum energy configurations for contact binary asteroids at different levels of angular momentum are computed and discussed. We find distinct transitions between different configurations as the angular momentum of the system is increased. These indicate that rapidly rotating contact binary asteroids may seek out clearly different relative configurations than slowly rotating systems. We find a single end state of the systems prior to rotational fission, and distinct dynamical outcomes as a function of mass distribution and shape when the rotational fission limit is exceeded. Our theoretical results agree qualitatively with observed properties of near-Earth asteroids, and can be used to help explain the spin-rate barrier, contact binaries, and the observed morphology of most NEO binaries.

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1. Introduction

The environment near-Earth objects (NEOs) live in subject them to a variety of perturbations that can change their rotational angular momentum. These perturbations include slow changes over time due to torques arising from momentum transfer due to incident and reemitted solar photons (known as the YORP effect) to abrupt changes due to planetary flybys or impacts. Of particular interest is what effect changes in angular momentum may have on a contact binary asteroid. In the early 1980s (Farinella et al., 1981; Weidenschilling, 1981) a series of investigations were published that studied the minimum energy equilibrium configuration of asteroids, but these treated asteroids as fluidic bodies, an assumption that is now regarded as unrealistic for smaller asteroids. The more recent work by Holsapple (2001, 2004) has modified the concept of an asteroid figure of equilibrium to incorporate shear strength. Richardson has studied the minimum energy configurations of rubble piles consisting of equal sized spheres, and their response to rapid changes in spin rates and tidal forces from planetary flybys

(Richardson et al., 1998, 2005; Walsh and Richardson, 2006a). Roig has carried out similar analyses using rubble piles constructed of ellipsoidal components (Roig et al., 2003). Sharma has considered the response of ellipsoidal stress fields modeled as rubble piles to planetary flybys (Sharma et al., 2006). Washabaugh and Scheeres have studied the minimum energy configurations of ellipsoids with realistic material properties (Washabaugh and Scheeres, 2002). These studies have all focused on simple ellipsoidal shaped aggregates. However, many asteroids are known to have apparently separate components resting on each other, so called contact binary asteroids. We find that the minimum energy configurations and evolution of contact binary asteroids are much different than for aggregates forming an ellipsoidal shape. Our current study provides analytical insights into minimum energy contact binary configurations as a function of their total rotational angular momentum.

As the rotational angular momentum of a contact binary NEO changes, the configuration of the components of the NEO may also change once certain energetic thresholds are crossed. These reconfigurations can be global, meaning that major changes in the orientation of the body's components may occur, potentially exposing material previously contained in the

E-mail address: scheeres@umich.edu.

body interior and burying previously exposed material. If angular momentum continues to increase it is even possible for the body to fission, with some portions of it reaching orbital speeds and separating from the surface. Similar situations have been proposed recently as candidates for slowing the rotation of a rapidly spinning asteroid (Harris, 2002) and potentially spawning binary asteroids (Bottke et al., 2002; Scheeres et al., 2006; Walsh and Richardson, 2006b). Furthermore, recent computations of the effect of YORP torques on asteroids have indicated that this effect should be widespread and present in the NEO population and even in the main belt (Rubincam, 2000; Vokrouhlický and Čapek, 2002; Taylor et al., 2007; Lowry et al., 2007).

Related to these physical phenomenon, in this paper we consider a number of simple questions regarding what should happen to contact binary asteroids spun to high spin rates, explicitly modeling the asteroid not as a single body but as composed of at least two components resting on each other. The specific questions we ask include: What is the minimum energy configuration of two rigid bodies resting on each other? What is the necessary spin rate for a contact binary to become an orbital binary? What is the dynamical stability of an orbital binary formed by a fission process?

For our prototype model we choose a simple system comprised of a constant density ellipsoid and a sphere of a given radius. We choose this system for a number of reasons, mostly related to simplicity and definiteness of the results. Given the interesting behavior we have found for this simple system, reported in this paper, we intend to extend this analysis to more complex shapes in the future.

We find that a system composed of a sphere and ellipsoid resting on each other have definite transitions between stability states as the angular momentum is changed. Thus, this indicates that contact binary asteroids may change their major configurations as their spin states evolve. We do not discuss the mechanism by which they may initiate this change, but we observe that it is a commonly accepted physical principle that all systems, if allowed to, will seek out their energy minimizing configuration. For contact binary asteroids, such opportunities to shift to an energy minimizing configuration may include relatively small impacts and planetary flybys. We also find that the stability of an asteroid that undergoes fission into an orbital system depends strongly on the mass distribution between the two bodies and on the shape of the bodies, or in our case the shape of the ellipsoid.

The results we find for our simple model predict three distinct evolutionary pathways for asteroids composed of multiple components that fit with the observed morphology of asteroids and binary asteroids in the NEO population. To be specific, it provides a clear mechanism for asteroids with multiple components to reside in the vicinity of the spin-rate barrier (Pravec and Harris, 2000), predicts that certain contact binary asteroids may remain contact binary asteroids even if accelerated to fission, thus leading to a semi-permanent class of contact binary asteroids, and predicts that stable binary systems will preferentially have spherical primaries independent of the shape of the secondaries. Furthermore, the analysis predicts that contact bi-

aries will undergo fission at a much lower rotation rate than a spherical body with regolith on the surface, the usual model used in determining the stability of loose surface material on an asteroid.

The paper is organized as follows. Section 2 presents our physical model assumptions and justifications for its use. Section 3 presents the mathematical model based on our physical model and discusses the possible relative equilibrium states for the system. Section 4 considers the stability of these relative equilibrium states and how they change with system parameters and rotational angular momentum. Section 5 discusses the implications of our results for the evolution of NEAs subject to persistent rotational accelerations and how more realistic physical models may change our results based on our idealistic model. Section 6 provides a summary of our conclusions.

2. Physical model

There are several proposed mechanisms for the formation of binary asteroids in the near-Earth asteroid (NEA) population. These include formation via an impact event, formation during a tidal rupturing event during a close planetary passage, and formation due to “fission” of an asteroid due to an increased spin rate resulting from either a close planetary passage or YORP torques. Some of these mechanisms are reviewed in Merline et al. (2002), Scheeres et al. (2002), Bottke et al. (2002), Richardson and Walsh (2006) and interpreted in terms of the binary asteroid system 1999 KW4 in Scheeres et al. (2006). Each of these formation mechanisms will have implications for the resultant binary asteroid systems. In this paper we apply recent theoretical work on the stability of ideal binary asteroid systems (Scheeres, 2002b, 2006) to derive some constraints on binary asteroid systems produced by “fission” and their likely properties.

We define asteroid fission to occur when the asteroid spin rate (angular momentum) is increased to the point where components of the body are no longer resting on each other, but instead have entered orbit about each other. A simple example is in order. Consider two spheres of equal density ρ , but different radii, resting on each other, the entire system rotating about its center of mass (Fig. 1). First let one sphere have a non-zero radius and mass while the other is a material point resting on its surface (at the equator, we presume). Then if the system rotates faster than $\sqrt{4\pi G\rho/3}$ (where G is the gravitational constant), equal to a 2.4-h rotation period for a density of 2 g/cm^3 , the gravitational and centripetal accelerations balance and the two components will enter a mutual orbit and no longer rest on each other. If instead, the two spheres have equal radius and equal mass and rest on each other, then if the system rotates faster than $\sqrt{\pi G\rho/3}$, a 4.8-h rotation period for the above case, they will start to orbit each other. Obviously, the two equal sized masses will fission well before the material point will leave the surface. Each instance, however, should be viewed as an occurrence of asteroid fission.

There are two main mechanisms by which this increase in angular momentum can occur for NEOs: spin-up by planetary flyby and spin-up by YORP. In the near-Earth population im-

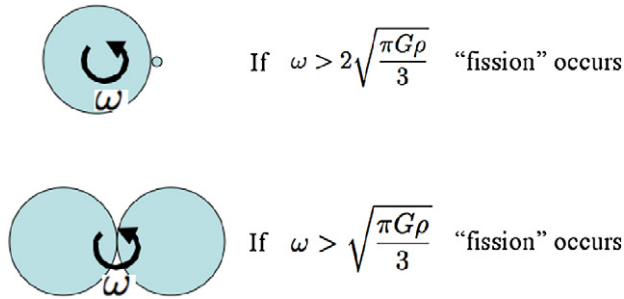


Fig. 1. Fission limits for spherical bodies with different masses.

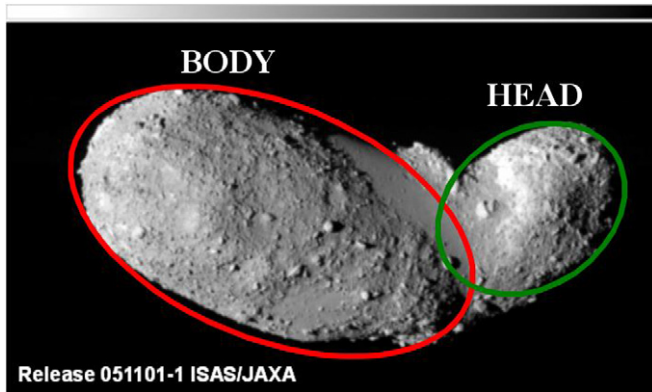


Fig. 2. Itokawa with its two components highlighted, following Demura et al. (2006). Modified from an image taken from Japanese Aerospace Exploration Agency (2006).

pacts play a diminished role, although they are very important for main belt asteroids. Repeated planetary flybys will, on average, cause asteroid rotation rates to increase, and could eventually lead to an asteroid having enough angular momentum to fission (Scheeres et al., 2004). Similarly, a small asteroid that is being consistently spun-up by YORP torques would naturally arrive at a rotation rate that would lead to fissioning (Bottke et al., 2002). We specifically focus on the effect of YORP on rotational fission, although our results are independent of the actual physical process that leads to the spin-up under our rigid body hypothesis. For bodies modeled as true rubble piles a planetary encounter may have a distinctive signature as compared to a gradual change in rotation rate as would occur from YORP.

Recent images of the Asteroid Itokawa (Fujiwara et al., 2006) as well as many results obtained from radar astrometry of asteroids (Benner et al., 2006) show that asteroids commonly have apparently distinct components that seem to rest on each other, so called contact binaries. The poster child of such contact binaries is Itokawa and its “head” and “body,” as highlighted in Fig. 2. The ability of rubble pile asteroids such as Itokawa to retain a characteristic shape in the presence of gravitational attraction of the other parts of the asteroid implies that they must have some shear strength. The ability of a rubble pile to sustain internal stresses is well established by the work of Holsapple (2001, 2004), and thus there is no controversy with a body being a contact binary and for the different components of that body being rubble piles.

If two rubble piles can retain their characteristic, distinct shapes while in contact with each other, then they can also re-

tain their characteristic shapes if shifted relative to each other or if they have slow impacts on the order of orbital speeds. Indeed, the stresses placed across the two components when in orbit will be lower than the stresses placed across them when they are lying on each other in extreme close proximity.

Based on these observations, it is feasible to treat the geometrically separate components of observed asteroids as coherent structures, and model them as rigid bodies as a first approximation. Of course, such a model neglects the rubble pile structure of the components—but is supported by the basic observations that asteroids retain their coherent structure even when in contact with each other. On the other hand, if one rejects the rubble pile hypothesis for these NEA, or assumes that the components of a contact binary are indeed monolithic, then the rigid body assumption is uncontroversial. Given the above arguments, we apply our model of contact binary asteroids as composed of rigid bodies of finite size resting on each other. This current discussion only uses ideal shapes, such as spheres and ellipsoids, in order to make general observations. For real asteroid systems it is expected that local topography will play an extremely important role in constraining and controlling asteroid evolution.

We now define our physical model. Consider a binary system consisting of a sphere of radius R and an ellipsoid with semi axes $\alpha_1 \geq \alpha_2 \geq \alpha_3$. We define the mass fraction of the system to be $\nu = M_s / (M_s + M_e)$, $0 \leq \nu \leq 1$, where M_s is the mass of the sphere and M_e is the mass of the ellipsoid. If the components have equal density, the radius of the sphere is $R = (\alpha_1 \alpha_2 \alpha_3)^{1/3} (\nu / (1 - \nu))^{1/3}$. We make no assumption about the relative size of the sphere and ellipsoid, and our discussion is relevant for the entire range of a small sphere and large ellipsoid to a large sphere and small ellipsoid.

Due to the symmetry of the ellipsoid, the relative equilibria that can exist between these two bodies can be easily enumerated. For a sphere and ellipsoid resting on each other and uniformly rotating, the sphere generally must be located along a principal axis of the ellipsoid and the system rotating about one of the principal axes of the ellipsoid. To analytically compute the stability of these different configurations is difficult. However, since we have identified a discrete set of equilibria it is possible for us to delineate the minimum energy configuration that such a system can have. We consider the energy of different configurations at a constant value of angular momentum to find the minimum energy configuration. Then we study how this minimum energy configuration changes as the angular momentum is increased.

At an angular momentum near zero the minimum energy configuration has the sphere resting on the α_3 axis of the ellipsoid for any $\alpha_1 \geq \alpha_2 > \alpha_3$. This remains the minimum energy configuration when the system rotates slowly. Conversely, for large enough angular momentum (but usually less than the value for which the components orbit each other) the minimum energy configuration consists of the sphere lying along the α_1 axis of the ellipsoid, the entire system rotating about the α_3 axis of the ellipsoid. These transitions are illustrated in Fig. 3. This result holds for most values of ν and all $\alpha_1 > \alpha_2 > \alpha_3$.

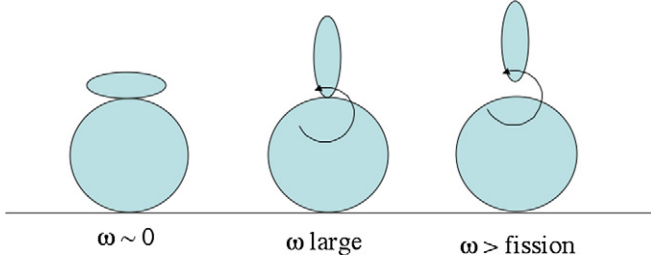


Fig. 3. Minimum energy configurations at different system spin rates ω .

For systems with $\nu \ll 1$ (a small sphere on a large ellipsoid), these results are easy to imagine. When not spinning the minimum potential point on the ellipsoid is always along the minimum axis. Thus small particles will preferentially move towards the polar regions. As the body rotates more rapidly, due to increases in its rotational angular momentum, the minimum energy point on the asteroid will shift at some point, in the limit always lying at the long ends of the spinning ellipsoid (Demura et al., 2006). The existence of these energetic transitions are independent of the mass fraction, and thus for systems with $\nu \sim 1$, ellipsoidal rocks on the equator of a rotating sphere will move from resting on their minimum axis for slow rotation rates towards standing on end as the angular momentum reaches a large enough value. For any of these cases, if the rotation rate increases to the fission limit, the components would naturally separate and the system would transition smoothly into an equilibrium orbital configuration. Such orbital configurations may be stable or unstable, depending on the mass fraction and shape of the ellipsoid, and are discussed later. These stability results can be directly related to possible evolution of the system and formation of binary asteroids.

3. Mathematical model

In the following we define the generic model described above, including the relevant relational equations for the system and stability conditions. In our discussion of the mutual potential, however, we initially take a more general approach.

3.1. Mutual and self-potentials

The heart of the current work is the mutual potential between two massive bodies. Thus we start with a general discussion of the total gravitational potential and then simplify it to the mutual potential of an ellipsoid–sphere system.

The total gravitational potential of a collection of mass $\mathcal{B}$ can be stated as the double integral of the Newtonian potential over itself (Scheeres, 2002b):

$$V = -\frac{1}{2} \int_{\mathcal{B}} \int_{\mathcal{B}} \frac{\mathcal{G} dm(\rho_1) dm(\rho_2)}{|\rho_1 - \rho_2|}, \quad (1)$$

here $\mathcal{B}$ represents the mass distribution, dm represents a mass element of the body, and ρ_i represents the position of the i th mass element. For our model of a contact binary asteroid we note that the total potential can be arbitrarily split into multiple components with the resultant potential consisting of the self

potential of each of the components plus all of the mutual potentials between the components.

Specifically, let us split the mass distribution into two disjoint collections of mass, denoted as $\mathcal{B}_1$ and $\mathcal{B}_2$, and assume that these bodies have centers of mass $\mathbf{r}_1$ and $\mathbf{r}_2$, respectively. Then the total potential can be split into three terms (Scheeres, 2004a):

$$V = V_{11} + V_{22} + V_{12}, \quad (2)$$

$$V_{ii} = -\frac{1}{2} \int_{\mathcal{B}_i} \int_{\mathcal{B}_i} \frac{\mathcal{G} dm(\rho_1) dm(\rho_2)}{|\rho_1 - \rho_2|}, \quad (3)$$

$$V_{12} = - \int_{\mathcal{B}_1} \int_{\mathcal{B}_2} \frac{\mathcal{G} dm(\rho_1) dm(\rho_2)}{|\mathbf{r} + \rho_1 - \rho_2|}, \quad (4)$$

where $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$ is the relative vector between the two-body centers of mass. Implicit in this formula are the relative orientation of the two bodies. It is important to note that when the total energy of the system is computed the two self-potentials must, for the most general case, be included, meaning that excess energy in the system can always be absorbed into the self-potentials of the system through changes in the shape, or conversely that changes in shape can make additional energy available to the system dynamics. While we do not consider such changes here, it is important to keep these possibilities in mind for future studies. While the potential can be split between portions of the mass distribution arbitrarily, we also assume that each such component is rigid but that the two components are not tied to each other. This is, mathematically, our contact binary assumption.

Now we introduce the “Sphere Restriction Assumption” that one body is a sphere. A quadrature over the sphere can be immediately performed and a much simpler single integral can be considered

$$V_{12} = -\mathcal{G} M_s \int_{\mathcal{B}_e} \frac{dm(\rho)}{|\mathbf{r} + \rho|}, \quad (5)$$

where the s subscript stands for the sphere and the e subscript stands for the general body, and M_s denotes the total mass of the sphere. This equation is proportional to the gravitational potential of the arbitrary body. We assume the quadrature takes place in a coordinate frame fixed in the general body at its center of mass. The mutual potential then becomes

$$V_{12}(\mathbf{r}) = -\mathcal{G} M_s M_e \tilde{U}(\mathbf{r}), \quad (6)$$

$$\tilde{U}(\mathbf{r}) = \frac{1}{M_e} \int_{\mathcal{B}_e} \frac{dm(\rho)}{|\mathbf{r} + \rho|}. \quad (7)$$

The function $\tilde{U}$ is the unit mass force potential of the general body and under a constant density assumption is a function of the body’s geometry only.

Now assume that the non-spherical body is a constant density, tri-axial ellipsoid. In Scheeres (1994) we find the general equations for the gravitational potential of an ellipsoid applied to an orbital problem [originally derived using Ivory’s theorem, cf. Danby (1989)], now stated in a coordinate frame taken along

the ellipsoid's principal axes of inertia:

$$\tilde{U} = \frac{3}{4} \int_{\lambda}^{\infty} \phi(\mathbf{r}, u) \frac{du}{\Delta(u)}, \quad (8)$$

$$\phi(\mathbf{r}, u) = 1 - \frac{r_1^2}{\alpha_1^2 + u} - \frac{r_2^2}{\alpha_2^2 + u} - \frac{r_3^2}{\alpha_3^2 + u}, \quad (9)$$

$$\Delta(u) = \sqrt{(\alpha_1^2 + u)(\alpha_2^2 + u)(\alpha_3^2 + u)}, \quad (10)$$

where $0 < \alpha_3 \leq \alpha_2 \leq \alpha_1$, and λ satisfies the implicit equation $\phi(\mathbf{r}, \lambda) = 0$, and is the largest positive, real root of this equation. The r_1 coordinate is along the longest axis of the body (minimum moment of inertia), while the r_3 coordinate is along the shortest axis of the body (maximum moment of inertia). The principal moments of inertia for this body are

$$\mathcal{I}_{11} = \frac{1}{5}(\alpha_2^2 + \alpha_3^2), \quad (11)$$

$$\mathcal{I}_{22} = \frac{1}{5}(\alpha_1^2 + \alpha_3^2), \quad (12)$$

$$\mathcal{I}_{33} = \frac{1}{5}(\alpha_1^2 + \alpha_2^2). \quad (13)$$

3.2. Energy and angular momentum

Essential quantities for the evolution of a dynamical system are its total energy and angular momentum, involving both rotational and translational quantities. The angular momentum is computed as

$$\mathbf{K} = \mathbf{A}_e \cdot \mathcal{I}_e \cdot \boldsymbol{\Omega}_e + \mathbf{A}_s \cdot \mathcal{I}_s \cdot \boldsymbol{\Omega}_s + \frac{M_e M_s}{M_e + M_s} \mathbf{r} \times \dot{\mathbf{r}}, \quad (14)$$

where $\mathbf{A}_i$ is the attitude transformation from the i th body fixed frame to an inertial frame, $\mathcal{I}_i$ is the inertia dyad, $\boldsymbol{\Omega}_i$ is the angular velocity, $\mathbf{r}$ is the relative position and $\dot{\mathbf{r}}$ is the time rate of change of the relative position with respect to an inertial frame.

The total energy of the system is

$$E = \frac{1}{2} \boldsymbol{\Omega}_e \cdot \mathcal{I}_e \cdot \boldsymbol{\Omega}_e + \frac{1}{2} \boldsymbol{\Omega}_s \cdot \mathcal{I}_s \cdot \boldsymbol{\Omega}_s + \frac{1}{2} \frac{M_e M_s}{M_e + M_s} \dot{\mathbf{r}} \cdot \dot{\mathbf{r}} + V_{ee} + V_{ss} - \mathcal{G} M_e M_s \tilde{U}(\mathbf{r}, \mathbf{A}_e, \mathbf{A}_s). \quad (15)$$

We note that the self-potential is properly retained in these general equations, as changes in the body shapes/self-potentials can cause changes in the dynamical parameters, subject to the angular momentum constraint. In Scheeres (2004a) the contribution of self-potentials to the overall dynamical evolution of a system are explored in more detail and with more rigor. For the remainder of the paper, however, we will suppress the self-potentials and not consider them. This is equivalent to assuming that the bodies are rigid and do not deform.

Distinctions must be made between the angular momentum and energy available for evolution of the system when the bodies are in contact and when they are not. Specifically, if the two bodies are not in contact the rotational energy and angular momentum of the sphere cannot be changed, and can be removed from the system entirely. For this situation we define the “free”

energy and angular momentum, computed now as

$$\mathbf{K}_F = \mathbf{A}_e \cdot \mathcal{I}_e \cdot \boldsymbol{\Omega}_e + \frac{M_e M_s}{M_e + M_s} \mathbf{r} \times \dot{\mathbf{r}}, \quad (16)$$

$$E_F = \frac{1}{2} \boldsymbol{\Omega}_e \cdot \mathcal{I}_e \cdot \boldsymbol{\Omega}_e + \frac{1}{2} \frac{M_e M_s}{M_e + M_s} \dot{\mathbf{r}} \cdot \dot{\mathbf{r}} - \mathcal{G} M_e M_s \tilde{U}(\mathbf{r}, \mathbf{A}_e). \quad (17)$$

When we consider the situation when the two bodies are touching each other, we have additional physics to consider, related to the surface friction between the bodies and the supporting normal forces. The two limiting extremes are a frictionless interaction or a no-slip friction model between the bodies. Each of these cases can be easily handled analytically, although for intermediate values of friction the analytics become more difficult and require explicit computation of the contact forces. For the frictionless case the sphere again is removed from consideration for the total energy and angular momentum as there are no torques that can act on it. For this case the above free energy and angular momentum are the appropriate formulae to use. Such a situation is not realistic, however.

The case of a no-slip friction model is more realistic, given the rough surfaces of asteroids, and implies that rolling without slipping occurs between the two bodies. As the bodies move relative to each other they again must conserve angular momentum. Energy will, in general, be dissipated when such rolling motion between the resting bodies occurs.

3.3. Existence of relative equilibria

Given our simple model it is possible to delineate all possible relative equilibria in the system. We consider two types, orbital and resting equilibria.

3.3.1. Orbital relative equilibria

Orbital relative equilibria have been considered extensively in Scheeres (2006) and can be easily discussed. For an ellipsoid–sphere to be in an orbital relative equilibrium all that is required is that the sphere be located along a principal moment of inertia of the ellipsoid and the entire system rotate with a specific rate about a different principal moment of inertia of the ellipsoid. Given such a configuration the necessary spin rate for the system to remain in this configuration is computed from

$$\omega^2(r, \alpha_i) = \frac{3}{2} \mathcal{G}(M_s + M_e) \int_{r^2 - \alpha_i^2}^{\infty} \frac{du}{(\alpha_i^2 + u) \Delta(u)}, \quad (18)$$

where ω is the spin rate, α_i is the semi-major axis along which the sphere lies, and r is the distance between the two mass centers. Note that the rotation rate is independent of which other axis the system rotates around, so long as it is not the i axis. Thus we note that there are 6 distinct equilibrium configurations the system can lie in. Most are always unstable, however, a topic we discuss later. We note that these orbital relative equilibria can be extended to non-symmetric bodies as well (Scheeres, 2006).

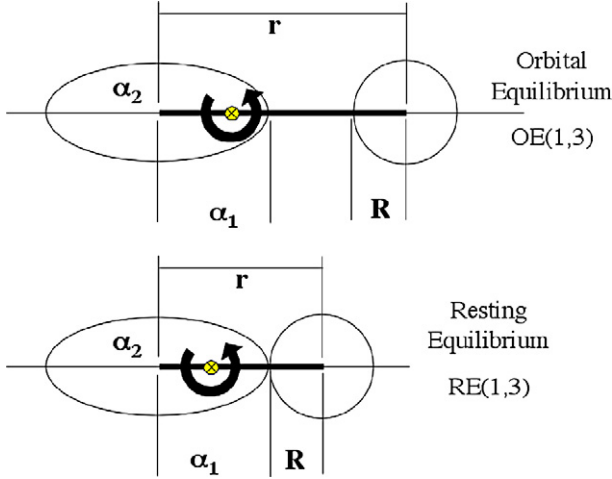


Fig. 4. An OE(1, 3) orbital equilibrium and a RE(1, 3) resting equilibrium.

To denote the different, distinct possible orbital relative equilibria we define the notation OE(i, j), where OE denotes orbital relative equilibria, i denotes the axis of the ellipsoid along which the two bodies lie, and j denotes the axis of the ellipsoid about which the system rotates. We note that we cannot have $i = j$. For example OE(1, 3) denotes that the sphere is aligned along the α_1 axis with the entire system rotating about the α_3 axis at a rate $\omega(r, \alpha_1)$ (see Fig. 4).

3.3.2. Resting relative equilibria

For a sphere and ellipsoid resting on each other to be in a relative equilibrium, the sphere must be resting on a principal axis of the ellipsoid and the system must be rotating about one of the principal axes of the ellipsoid. Generic resting equilibria exist due to the symmetry of the ellipsoidal gravity field about its principal planes. For some special combinations of rotation rate and shape additional surface resting equilibria may exist. For example, if the ellipsoid is a Jacobi ellipsoid (i.e., has the shape of a Jacobi ellipsoid and spins at the proper rotation rate) then every point on its surface is a resting equilibrium for a material point (Guibout and Scheeres, 2003). However, these situations are singular and we will not consider them here. Resting equilibria will also exist at bodies without symmetry, although the computation of the resting equilibrium locations will be more difficult. For our generic case, the mutual distance is $r = \alpha_i + R$, where R is the sphere radius. To be at rest the spin rate must either be about an axis $\neq i$ and less than $\omega(\alpha_i + R, \alpha_i)$, or the body can rotate about the axis i at an arbitrary rate. In either case the surfaces exert supporting forces on each other to maintain a stationary configuration. A similar surface equilibrium was considered in Guibout and Scheeres (2003), but in that case the sphere was assumed to be a massless particle. Should the spin rate about an axis not equal to i exceed $\omega(\alpha_i + R, \alpha_i)$ the surface relative equilibria will cease to exist.

Introducing a similar notation as in the orbital relative equilibrium case, we define the notation RE(i, j), where RE denotes resting equilibrium, i is the ellipsoid axis along which the sphere is resting, and j is the ellipsoid axis about which the system is rotating. Note that for the resting equilibrium we can

have $i = j$. Thus, for the given resting equilibrium RE(i, j) the total energy and angular momentum are specified as K_{ij} and E_{ij} , each of these being a function of system rotation rate Ω . See Fig. 4 for a graphic of an RE(1, 3) resting equilibrium.

The energy and angular momentum of the system at a relative equilibrium can be computed as follows. If we assume the sphere is located along axis i and rotates about axis j , where $i \neq j$ and $k \neq i, j$, then we compute

$$K_{ij} = \left[\frac{2}{5} M_s R^2 + \frac{1}{5} M_e (\alpha_i^2 + \alpha_k^2) + \frac{M_s M_e}{M_s + M_e} (\alpha_i + R)^2 \right] \Omega, \quad (19)$$

$$E_{ij} = \frac{1}{2} \left[\frac{2}{5} M_s R^2 + \frac{1}{5} M_e (\alpha_i^2 + \alpha_k^2) + \frac{M_s M_e}{M_s + M_e} (\alpha_i + R)^2 \right] \Omega^2 - \mathcal{G} M_s M_e \tilde{U}(\alpha_i + R) \quad (20)$$

$$= \frac{1}{2} \frac{K_{ij}^2}{\left[\frac{2}{5} M_s R^2 + \frac{1}{5} M_e (\alpha_i^2 + \alpha_k^2) + \frac{M_s M_e}{M_s + M_e} (\alpha_i + R)^2 \right]} - \mathcal{G} M_s M_e \tilde{U}(\alpha_i + R). \quad (21)$$

If the body rests along axis i and the system rotates about that same axis we have slightly different results:

$$K_{ii} = \left[\frac{2}{5} M_s R^2 + \frac{1}{5} M_e (\alpha_j^2 + \alpha_k^2) \right] \Omega, \quad (22)$$

$$E_{ii} = \frac{1}{2} \left[\frac{2}{5} M_s R^2 + \frac{1}{5} M_e (\alpha_j^2 + \alpha_k^2) \right] \Omega^2 - \mathcal{G} M_s M_e \tilde{U}(\alpha_i + R) \quad (23)$$

$$= \frac{1}{2} \frac{K_{ii}^2}{\left[\frac{2}{5} M_s R^2 + \frac{1}{5} M_e (\alpha_j^2 + \alpha_k^2) \right]} - \mathcal{G} M_s M_e \tilde{U}(\alpha_i + R). \quad (24)$$

We note that we are only concerning ourselves with the angular momentum magnitude, and that we restate the energy as a function of angular momentum to allow for comparison of the energy of different configurations. The above expressions are for the resting configurations assuming rolling without slipping between the components. To change these to the frictionless sliding case we need only drop the moment of inertia of the sphere from these equations.

To analytically compute the stability of these different configurations is difficult, however since we have identified a discrete set of equilibria, it is possible for us to delineate the minimum energy configuration that such a system can have. We then make the argument that, in the presence of perturbations such as planetary flybys and impacts, we expect the system to preferentially seek out these minimum energy configurations over time. Note, we tacitly assume that the system can energetically transition from one configuration state to another; this assumption should be investigated in the future.

3.4. Normalization

Now we introduce a series of normalizations to reduce the number of free parameters. Specifically, we introduce normalizing quantities for the mass (M_n), length (R_n), and time ($1/\Omega_n$).

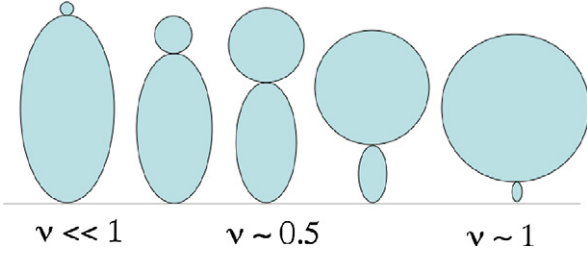


Fig. 5. Mass fraction variation for ellipsoids and spheres.

These are computed as

$$M_n = \frac{M_s M_e}{M_s + M_e}, \quad (25)$$

$$R_n = \left(\frac{3(M_s + M_e)}{4\pi\rho} \right)^{1/3}, \quad (26)$$

$$\Omega_n = \sqrt{\frac{4\pi}{3}} G\rho, \quad (27)$$

where ρ is the density of the two bodies, assumed equal. Given these normalizations we also find simple expressions for the normalized radius of the sphere and the maximum semi-major axis of the ellipsoid. To find these we assume that the total mass of the system is fixed, and that as the mass fraction parameter v is changed the sizes of the two bodies adjust. Because of the choice of normalizing radius we have a simple relationship between the mass fraction and the sphere and ellipsoid radii:

$$\bar{R} \equiv \frac{R}{R_n} \quad (28)$$

$$= v^{1/3}, \quad (29)$$

$$\bar{\alpha} \equiv \frac{\alpha_1}{R_n} \quad (30)$$

$$= \left(\frac{1-v}{\tilde{\alpha}_2 \tilde{\alpha}_3} \right)^{1/3}, \quad (31)$$

where $\tilde{\alpha}_2 = \alpha_2/\alpha_1$ and $\tilde{\alpha}_3 = \alpha_3/\alpha_1$. With these definitions we note that $\tilde{\alpha}_i = \bar{\alpha}$, $\tilde{\alpha}_2 \tilde{\alpha}_3 = \bar{\alpha}^2$ for $i = 1, 2, 3$, respectively. In Fig. 5 we show the relationship between ellipsoid and sphere size and the mass fraction v .

Normalizing the total energy and angular momentum then yields:

$$\bar{K}_{ij} = \left[\frac{2}{5} \frac{\bar{R}^2}{1-v} + \frac{1}{5} \frac{\tilde{\alpha}_i^2 + \tilde{\alpha}_k^2}{v} + (\tilde{\alpha}_i + \bar{R})^2 \right] \bar{\Omega}, \quad (32)$$

$$\bar{E}_{ij} = \frac{1}{2} \left[\frac{2}{5} \frac{\bar{R}^2}{1-v} + \frac{1}{5} \frac{\tilde{\alpha}_i^2 + \tilde{\alpha}_k^2}{v} + (\tilde{\alpha}_i + \bar{R})^2 \right] \bar{\Omega}^2 - \tilde{U}(\tilde{\alpha}_i + \bar{R}) \quad (33)$$

$$= \frac{1}{2} \frac{\bar{K}_{ij}^2}{\left[\frac{2}{5} \frac{\bar{R}^2}{1-v} + \frac{1}{5} \frac{\tilde{\alpha}_i^2 + \tilde{\alpha}_k^2}{v} + (\tilde{\alpha}_i + \bar{R})^2 \right]} - \tilde{U}(\tilde{\alpha}_i + \bar{R}), \quad (34)$$

where $\bar{\Omega} = \Omega/\Omega_n$.

In the normalized system as $v \rightarrow 0, 1$ the energy and angular momentum become arbitrarily large. This arises due to the mass normalization used, but is realistic as it denotes that the contribution of one of the bodies to these quantities becomes

negligible, or conversely that the remaining component appears to be immovable relative to the small particle. We shall see in the next section that other important limits for this system do not suffer a similar singularity, however.

3.5. Fission limits

A crucial item in the current analysis is the precise rotation rate at which a surface resting equilibrium transitions into an orbiting relative equilibrium. We call this the fission limit, and as mentioned earlier, it depends on the distribution of mass between the two bodies and their relative size. Using our previously defined relative equilibrium conditions we see that the fission limit equals $\omega(R + \alpha_i, \alpha_i)$. Introducing the normalization we formally define the fission limit as

$$\bar{\omega}_i(v, \tilde{\alpha}_2, \tilde{\alpha}_3) \equiv \omega(R + \alpha_i, \alpha_i)/\Omega_n, \quad (35)$$

where

$$\bar{\omega}_i^2 = \frac{3}{2} \int_{\lambda}^{\infty} \frac{du}{(\tilde{\alpha}_i^2 + u)\Delta(u)}, \quad (36)$$

$$\lambda = (\bar{R} + \tilde{\alpha}_i)^2 - \tilde{\alpha}_i^2, \quad (37)$$

$$\Delta(u) = \sqrt{(\tilde{\alpha}_1^2 + u)(\tilde{\alpha}_2^2 + u)(\tilde{\alpha}_3^2 + u)}. \quad (38)$$

We note again that the fission limit is independent of what axis the system rotates about, so long as it does not rotate about axis i . The integral in Eq. (36) can be evaluated in terms of Carlson elliptic integrals (Press et al., 1992).

This integral can be reduced to an especially simple form for the special case of two spheres resting on each other, $\tilde{\alpha}_1 = \tilde{\alpha}_2 = \tilde{\alpha}_3$. The fission limit formula simplifies to

$$\bar{\omega}^2(v) = \frac{3}{2} \int_{[v^{1/3} + (1-v)^{1/3}]^2 - (1-v)^{2/3}}^{\infty} \frac{du}{[(1-v)^{2/3} + u]^{5/2}} \quad (39)$$

$$= \frac{1}{[v^{1/3} + (1-v)^{1/3}]^3}. \quad (40)$$

It is simple to note that for the double sphere case $v = 0, 1$ yields a value of unity, which corresponds to the spin rate necessary for a mass-less particle on the surface of a sphere to achieve circular orbit. The minimum value of the relation for two spheres occurs for $v = 1/2$ and yields a fission spin rate $\bar{\omega}(1/2) = 1/2$ (see Fig. 1).

In Fig. 6 we plot the sphere case and a few sphere–ellipsoid cases for comparison, assuming the system fissions from the RE(1, 3) configuration. We note that when one of the bodies is an ellipsoid the fission curve is always lower than the double sphere case, and that the minimum is biased towards the ellipsoid-dominant side. Thus, bodies with an elongate primary will preferentially fission early, while bodies dominated by a spheroidal primary will fission at a higher rotation rate. For the ellipsoid dominant cases ($v \ll 1$) we note that the minimum rotation rate always occurs for the RE(1, 3) configuration, i.e., the sphere along the long axis of the ellipsoid. It is interesting

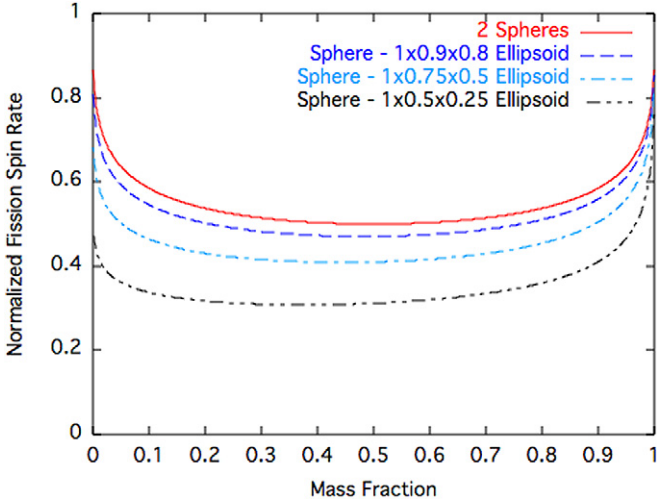


Fig. 6. Fission rates for binary systems from a RE(1, 3) configuration.

to note that these are not always the minimum angular momentum configurations for fission, a topic that will be discussed in Section 4.2.

We explicitly make note of the strong effect that shape and mass fraction have on the fission rate. For mass fractions in the range $0.1 \leq \nu \leq 0.9$ the fission rates are over 40% lower than the usually quoted rate of $\sim 8 \times 10^{-4}$ rad/s, corresponding to a rotation period of ~ 2.2 h, for the spin-rate barrier (Pravec and Harris, 2000). Thus, we see that if a given asteroid is a contact binary it could be at its fission limit for rotation periods longer than 3.7 h and up to 6.3 h for the examples given in Fig. 6. Consideration of these effects could change the interpretation of the spin rate/size distributions given in (Pravec and Harris, 2000).

4. Stability of relative equilibria

There are multiple definitions of stability that apply to binary systems such as considered here. We choose three specific types of stability for consideration. These are configuration stability, escape stability, and impact stability. If a system is configuration stable, then it will nominally remain in its current configuration, even accounting for dissipation of energy (which can often destabilize marginally stable configurations). A system is stable against escape if it is impossible for the system to evolve into an escaping trajectory under its internal dynamics alone. Conversely, a system that is unstable with respect to escape can suffer such an escape. Finally, a system is stable against impact if it is impossible for the two components to impact each other, and unstable with respect to impact if they can impact. In Scheeres (2002b, 2006) specific mathematical criterion for these stability computations are given, and will be applied below.

4.1. Stability of orbiting equilibria

We will first consider the stability of orbiting relative equilibrium, as this proves to be a somewhat simpler case.

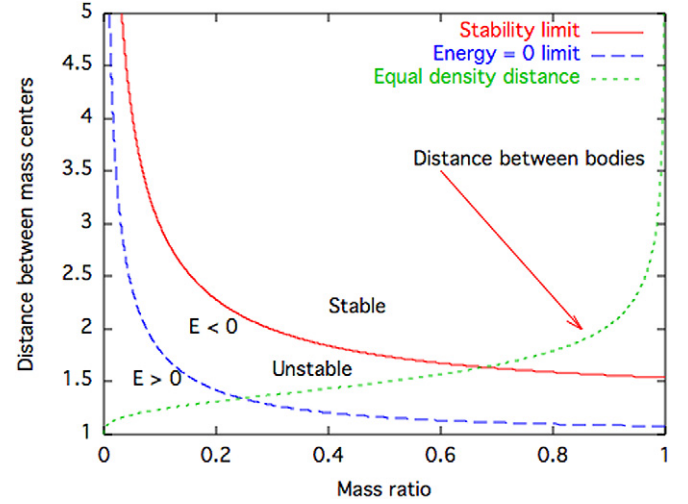


Fig. 7. Relative equilibrium stability diagram for an ellipsoid with semi-axes $1 \times 0.5 \times 0.25$. The distance between the bodies is normalized by the long axis of the ellipsoid.

4.1.1. Configuration stability

There is only one class of orbiting relative equilibria that can exhibit configuration stability in the presence of energy dissipation. These are the relative equilibria that place the sphere along the long axis of the ellipsoid, rotating about the shortest axis, or OE(1, 3) (see Fig. 3). While it is possible for OE(2, 3) configurations to be linearly stable, these are never energetically stable and hence will depart from this configuration under energy dissipation.

Systems in the OE(1, 3) configuration are common in natural and artificial systems. For natural satellites this is the well known 1:1 synchronous rotation and orbit state, the configuration most planetary moons have with their planet. Similarly, simple gravity gradient satellites also maintain this orientation. What is somewhat more surprising is that such OE(1, 3) configurations can actually be unstable in certain situations (Scheeres, 2004b). These situations only occur for close distances between the sphere and ellipsoid, and especially when the ellipsoid mass dominates. In fact, another well known result is that the relative equilibrium of a massless particle placed in a synchronous orbit along the long axis of an ellipsoid is unstable (Scheeres, 1994). In Figs. 7 and 8 we show stability diagrams for two sphere–ellipsoid systems with strong and mild elongation. We note that the qualitative form of the stable-unstable boundary line remains the same, even though the ellipsoids have very different parameters.

The interpretation of these plots is simple. Any point in the diagram can be an OE(1, 3) relative equilibrium, however the “equal density distance” is the separation between a sphere and an ellipsoid of equal density when resting on each other in a RE(1, 3) configuration. Thus, for equal density components we restrict ourselves to points above this line. The orbital equilibrium points above the “stability limit” all exhibit energetic stability and will be robust against energy dissipation and external perturbation (up to a limit). The points below the stability line, along with all other possible orbital equilibria, are unstable and cannot maintain their relative equilibrium configuration.

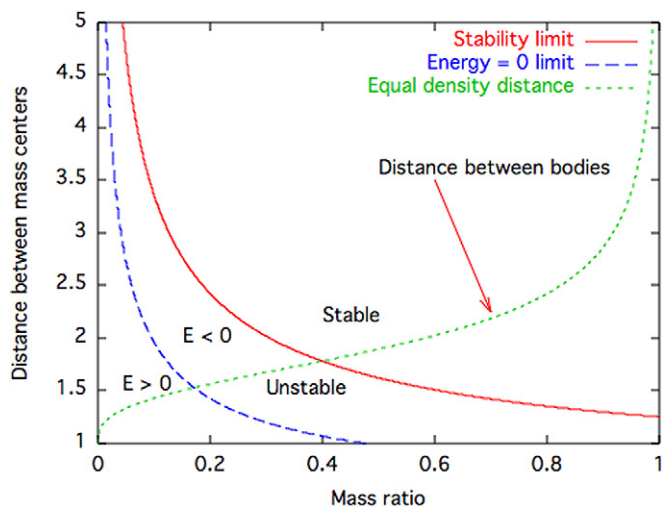


Fig. 8. Relative equilibrium stability diagram for an ellipsoid with semi-axes $1 \times 0.9 \times 0.8$. The distance between the bodies is normalized by the long axis of the ellipsoid.

The time scale for these systems to depart from their relative equilibrium when perturbed are extremely short, on the order of minutes to hours, and thus cannot form or persist for any length of time in these unstable configurations. Plotted in Fig. 9 is a trajectory of a sphere perturbed from its unstable equilibrium. The system has a mass fraction of 0.1, the ellipsoid has dimensions $1 \times 0.5 \times 0.25$, and the initial OE(1, 3) radius is 1.5. Note that there are two possible paths the orbit can evolve along, one that impacts with the ellipsoid and the other which falls away from the body. For this example the exterior orbit eventually escapes, as shown in Fig. 11. The Energy = 0 line in Figs. 7 and 8 denotes when the corresponding OE(1, 3) relative equilibrium has a positive or negative free energy. If the system has a positive free energy, the two components can escape from each other, if it has a negative free energy they remain bound to each other.

4.1.2. Impact limits

In Scheeres (2002b) a sufficient condition is given for a system to be stable against impact of its components. It is beyond the scope of this paper to apply that limit to the current discussion, and thus we will take a different approach to our discussion of this type of stability. When we consider the stable relative equilibria, it can be easily established that these are stable against impact (providing there is some separation between the bodies) as these are the true minimum energy points in phase space for a system with that angular momentum.

When the relative equilibrium is unstable, however, such constraints may no longer apply. Instead, the equilibrium point lies at an energy saddle, meaning that there are unstable trajectories leading away from the relative equilibrium both towards and away from the body. One of these is shown in Fig. 9. Thus, whenever a relative equilibrium is unstable there are natural pathways for the mutual orbit to evolve that can lead to impact between the bodies. Even when the system is close to stability, it is possible for these trajectories to impact with the surface. In Fig. 10 we show the unstable trajectories for the same el-

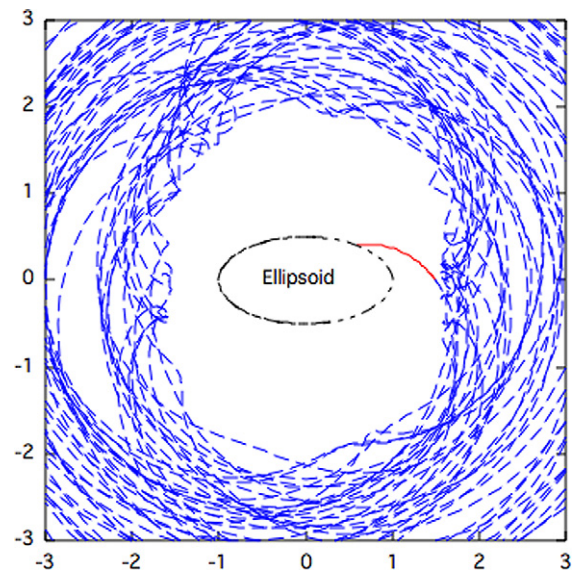


Fig. 9. Two possible paths of a sphere leaving an unstable relative equilibrium. The solid (inner) line impacts the surface while the dashed (outer) line eventually escapes. The trajectory is plotted in the central ellipsoid fixed frame. The ellipsoid has semi-major axes $1 \times 0.5 \times 0.25$.

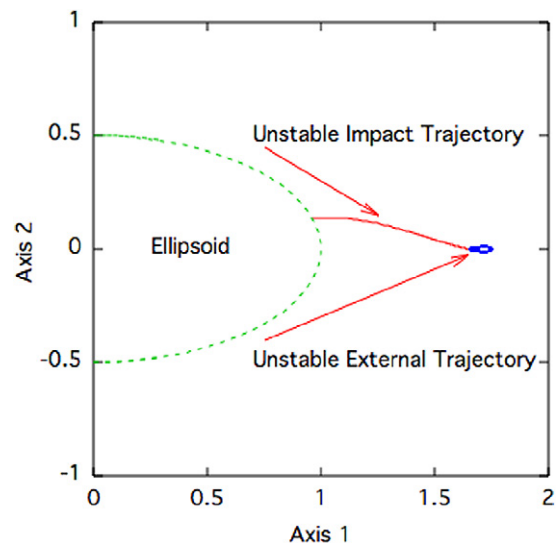


Fig. 10. Unstable trajectories of a relative equilibrium close to stability. The interior trajectory still impacts the surface while the exterior trajectory is trapped within a small region. The trajectory is plotted in the central ellipsoid fixed frame. The ellipsoid has semi-major axes $1 \times 0.5 \times 0.25$.

lipoid as in Fig. 9 but with a relative equilibrium defined by a mass fraction of 0.6 and a separation between the body centers of 1.65. This system is very close to being stable, as can be seen in the stability diagram (Fig. 7). The exterior unstable trajectory for this equilibrium point is trapped close to the equilibrium point with a very small oscillation, however the interior trajectory still impacts with the surface. If the system's relative distance is increased to above the stability line, the impacting trajectory no longer exists.

In addition to the dynamical pathways to impact, simple considerations of body shape and geometry can also influence

impact. Thus, when dealing with realistic bodies with asymmetries, perturbations, and surfaces with protuberances jutting off the surface the likelihood for bodies in a close relative equilibrium to strike each other will increase.

4.1.3. Escape limits

If a relative equilibrium starts deep enough in the unstable zone (small ν and small r), we see that the free energy of the system can be positive. As established in Scheeres (2002a, 2002b) this can lead to a disruption of the binary system via mutual escape. Simulations started in this region of the stability diagram clearly show that escape is a frequent, if not assured, outcome. It is important to note that when escape occurs, angular momentum and energy is carried away by the mutual orbit leaving the now separate bodies with slower spin rates. Thus, a system in this region spun to a rate where fission can occur will most likely have its rotation rate decreased significantly if disruption occurs. In Scheeres (2004a) explicit bounds on the final rotation rate of a system that fissions and has a mutual escape are given. Fig. 11 shows the eccentricity and rotation period of the escape trajectory from Fig. 9 to provide a concrete example of this phenomenon. As the total system free energy gets closer to zero, but remains positive, the final rotation rate of a disrupted system becomes smaller.

4.2. Energetic stability of resting equilibria

In the above we tacitly assumed that the relative equilibria formed by a system rotating to fission were generally in a RE(1, 3) configuration which leads naturally to an OE(1, 3) orbital equilibrium. In this section we will provide support for this scenario, in addition to raising a number of other issues related to the energetics of bodies resting on each other.

This section is a generalization of the study performed in Guibout and Scheeres (2003) who studied the stable locations of an infinitesimal particle on the surface of a rotating ellipsoid. That study found that as different rotation rates were considered, the preferred resting place of particles would shift from the minimum axis to the maximum axis. The primary of the binary Asteroid 1999 KW4 experiences such a situation, with the equator being the minimum potential on that body (Scheeres et al., 2006). Those situations can be directly applied to the distribution of regolith on the surface of the body. The current paper unifies those previous discussions by considering the minimum energy resting configuration of the general case of a sphere resting on an ellipsoid with an arbitrary mass fraction.

The dynamics of a system comprised of two bodies that are free to roll on each other is complex. It is of interest to study this general dynamical problem in more detail in the future, but we side-step that issue in the current paper by focusing instead on the minimum energy configurations of the system and fission limits as a function of angular momentum.

Our approach for determining the stability of a given resting configuration is simple. For a sphere–ellipsoid system at a given value of angular momentum less than the fission limit, we can explicitly delineate all possible resting relative equilibria that can occur. By computing the energy of each of these

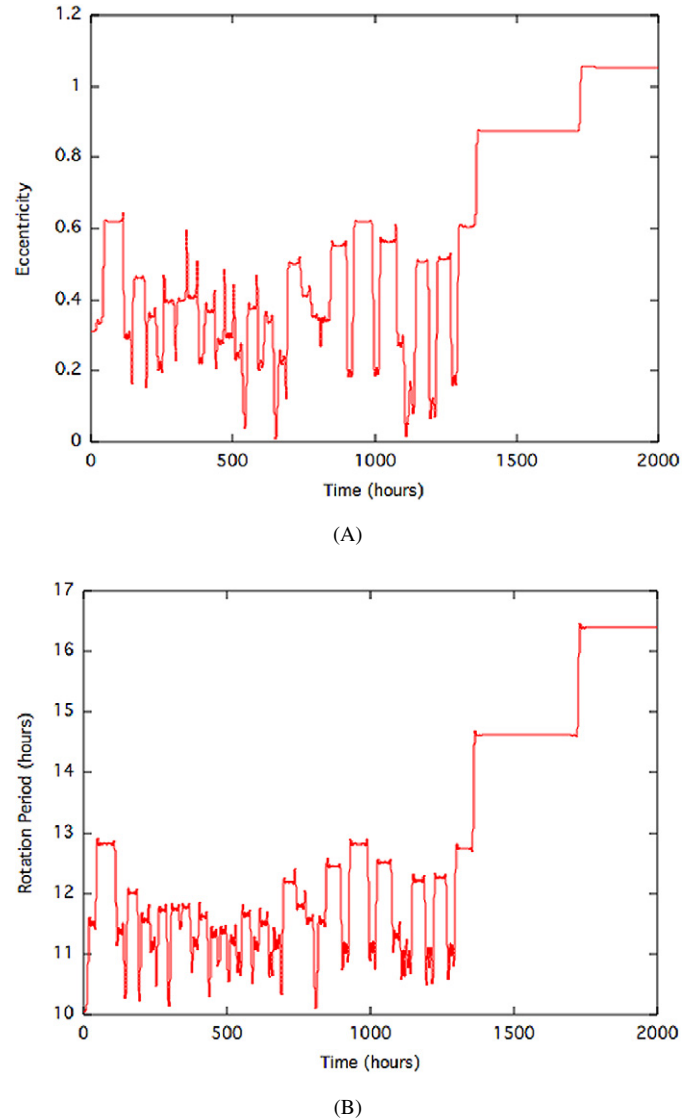


Fig. 11. (A) The eccentricity of the escaping trajectory from Fig. 9 over a 2000-h period. The system has disrupted when the eccentricity exceeds unity for an extended period of time. The initial eccentricity is for the orbital relative equilibrium and is non-zero, as explained in Scheeres et al. (2006). (B) The rotation period of the ellipsoid for this system. Note that disruption leaves the ellipsoid spinning with a much slower rotation rate than it initially had.

resting equilibria we can directly find the minimum energy configuration of the system, which should be the one it seeks out in the presence of external disturbances. As the system, by definition, does not have sufficient angular momentum to sustain a stable orbiting configuration these contact configurations are the only possible candidates for the minimum energy configuration. Once we find the one with minimum energy we have then found the “lowest point” that the system can achieve at its current value of angular momentum.

It is important to note that under an increasing angular momentum, due to YORP for example, a system is not compelled to seek out a new minimum energy configuration once it exists—much like a surface slope beyond its angle of repose need not fail immediately. The configuration will be susceptible to transition to this new minimum, however, in the presence

of an external disturbance such as a relatively distant planetary flyby, small impact, or the failure of some smaller region on the asteroid. The likely situation would be for the system to be placed in a super-critical energetic configuration which is no longer a minimum energy configuration either by increasing or decreasing its rotation rate. When in such a super-critical state, a change in the configuration can be triggered by a smaller event, similar to the onset of a landslide or avalanche.

To evaluate these minimum energy states we consider a given ellipsoid–sphere system with a mass fraction ν and compute the minimum energy configurations as the total angular momentum of the system varies from zero to a maximum, defined by the maximum fission limits along the different axes of the ellipsoid. We do not only choose the long-axis fission limit as the upper limit for angular momentum, as in some cases a different resting axis can remain stable beyond this limit. The discussion becomes more complex once we view the angular momentum as being the independent parameter that increases instead of the rotation rate. For the rotation rate alone, it is easy to show that the long-axis configuration (RE(1, 3)) always has fission at the lowest rate. When considering angular momentum, however, it is possible for fission to occur at a lower angular momentum if the system has the appropriate resting equilibrium configuration. We shall see that these other cases are unstable and that almost all sphere–ellipsoid systems that fission will have the RE(1, 3) configuration as their minimum energy state prior to fission.

If angular momentum is near zero, as is classically known (MacMillan, 1958) the minimum potential energy resting configuration is for the ellipsoid's smallest axis to be aligned with the sphere's center. Thus, if we want to track the evolution of the minimum energy configurations starting from $K \sim 0$, we note that the RE(3, 2) or RE(3, 3) configuration is always the minimum. Recall that the first number is the axis which the sphere rests on and the second number is the axis about which the system rotates. In general, as the angular momentum magnitude is increased the system's minimum energy configuration will either transition directly to a RE(1, 3) configuration, or may sometimes pass through a RE(2, 3) configuration before it achieves a RE(1, 3) configuration. Figs. 12–14 show energetic stability transition diagrams for some nominal example shapes.

There are some interesting features of these plots that should be explained. First, in the transition diagrams the unity value of angular momentum is equal to the RE(1, 3) fission limit angular momentum for that value of ν . If the (1, 3) minimum energy transition line is less than 1, then the system's minimum energy configuration when approaching fission will be the (1, 3) configuration, which can be continued to the OE(1, 3) configuration should the angular momentum increase beyond the fission limit. Note that the resulting orbital relative equilibrium is not necessarily stable or minimum energy, that condition must be checked from the system's stability diagram, such as given in Figs. 7 or 8.

For elongate bodies with $\nu \sim 1$ (elongate boulders on a spherical primary) we note that the transition to (1, 3) may occur beyond the (1, 3) fission limits (see Fig. 12). A body in this situation would remain in its minimum energy RE(3, 2) con-

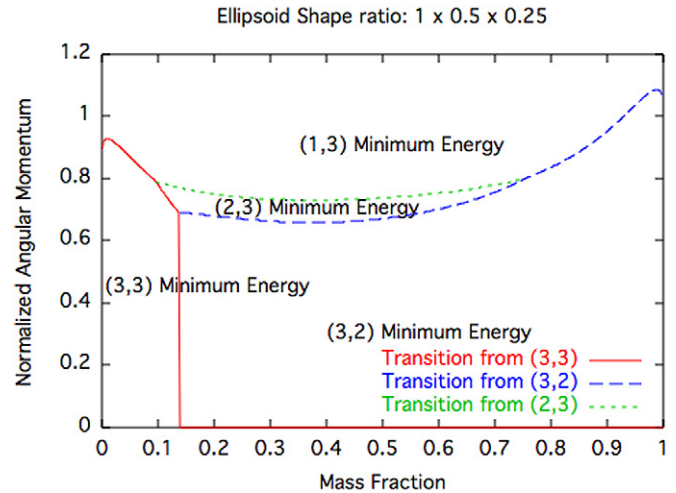


Fig. 12. Energetic stability transitions as a function of mass fraction and angular momentum for a sphere–ellipsoid system with ellipsoid axes $1 \times 0.5 \times 0.25$.

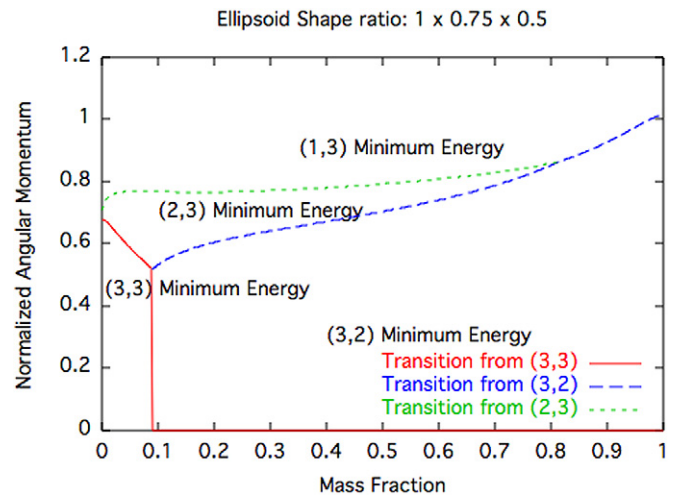


Fig. 13. Energetic stability transitions as a function of mass fraction and angular momentum for a sphere–ellipsoid system with ellipsoid axes $1 \times 0.75 \times 0.5$.

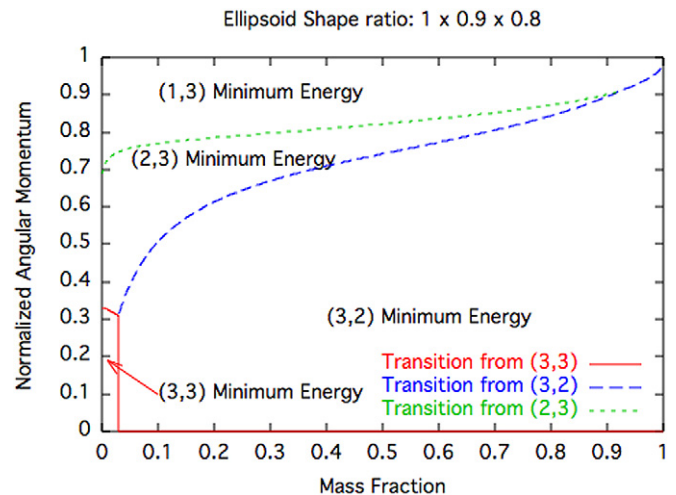


Fig. 14. Energetic stability transitions as a function of mass fraction and angular momentum for a sphere–ellipsoid system with ellipsoid axes $1 \times 0.9 \times 0.8$.

figuration up to its fission limit. Once fission occurs it enters into an OE(3, 2) orbital equilibrium which is unstable and out of which it would tumble. Under energy dissipation it could be trapped into a stable OE(1, 3) configuration, and would have a positive separation distance due to the excess angular momentum. As the ellipsoid becomes less elongate we note that the interval for which this behavior occurs shrinks (Fig. 13) and disappears (Fig. 14).

We note that there are cases where one of the other configurations, RE(3, 2) or RE(2, 3), can fission at angular momenta less than the long-axis limit. Should the system survive to fission in that configuration, however, the resulting orbit equilibrium would be unstable and could not remain in place. At these lower values of angular momenta the system cannot reside in an OE(1, 3) configuration and thus the system would reimpact. In general, for these situations the angular momentum is great enough so that the (1, 3) configuration is already the minimum energy configuration and these bodies would tend to evolve towards this configuration. Such a drastic change in configuration would reduce the spin rate of the system, could mix and shake the system dramatically, but could not disrupt it as the energies and speeds involved here are very small. Such a change could also change the net effect of YORP acting on the body and might even cause its rotation rate to evolve in the opposite direction, such as has been hypothesized for the Asteroid Itokawa (Scheeres et al., 2007). Specific simulation of such transitions would be of interest to perform in the future.

It is on the results described in this section that we base our claim that contact binary asteroids spun to fission will naturally seek out the RE(1, 3) configuration when approaching the fission limit. If rotational acceleration continues to act, then these RE(1, 3) configurations naturally transition into the OE(1, 3) orbital relative equilibrium. Their subsequent fate depends on the ellipsoid shape and the mass fraction of the system, and can be determined using the stability diagrams in Figs. 7 and 8.

5. Discussions and hypotheses

In the following we apply the above observations to the evolution of a contact binary asteroid subject to YORP accelerations and decelerations. We do not consider how the contact binary was created, but rely on the well established fact that asteroid shapes consistent with contact binary bodies are ubiquitous in the NEO population (Benner et al., 2006), as are binaries (Margot et al., 2002) and slow rotators (Harris, 2002). We propose that the combination of YORP and simple considerations of energy, angular momentum, and the dynamical stability of relative equilibria can provide a heuristically consistent picture of NEO evolution. Future studies will focus on specific issues that need to be tested in order to validate this as a component to an overall theory of NEO physical evolution.

Applying these issues together we identify three main evolutionary paths in this model that correspond to qualitative observed features of the NEO population. These are spin regulation by mass shedding, a contact binary cycle that does not allow a system to easily shed mass or evolve into a stable binary, and creation of stable binary asteroids.

5.1. Spin regulation

The first situation we consider is what occurs when a contact binary with an elongate primary ($\nu \ll 1$) reaches its fission rate. Such systems fall into the left-hand side of the stability diagrams in Figs. 7 and 8; when fission occurs these OE(1, 3) equilibria are unstable and have a positive free energy. Thus, the secondaries are susceptible to ejection from the system, which would decrease the total spin energy and angular momentum of the remaining primary (Fig. 15). Such a system would only approach the “spin-rate barrier” described in Pravec and Harris (2000) due to the lower rotation rate needed to fission. However, following ejection the primary would be subject again to YORP torques which may eventually cause an acceleration of the system. Depending on the largest block on the surface and the remaining primary shape, the system may go through such a cycle again, shedding mass and angular momentum. A body constructed such as Itokawa (without its head) could, in this way, first shed its largest boulder, and then progressively shed smaller boulders until it achieved a mass fraction and shape that allowed it to fall into one of the other cycles. Note, Harris (2002) first suggested a form of this spin regulation cycle, including comments on the total energy of the body. In this paper we formalize this cycle and tie it into a range of other possibilities.

It is interesting to note that the fission rate for bodies with elongate primaries is lower than fission rates for more spherical primaries, as seen in Fig. 6. Also, we note that the minimum fission rates as a function of mass fraction also coincide with the unstable region of the stability diagrams, cutting across both the positive and negative energy transition line. For the positive energy contact binaries that eventually escape the fission rate can be as low as 35% of the ideal limit, meaning that many of the rotators in the NEA population may actually be at their fission limit for rotation periods up to 6 hours or longer if they are contact binaries, far from the previously published fission rates. Thus, such shape morphology effects could be masking the fact that a larger fraction of NEAs are at their fission limit.

Another issue can be raised as relates to the existence of extremely slow rotators in the asteroid population. While the ability of a mutually escaping binary system to result in a slowly rotating parent has been clearly established in the literature (Harris, 2002; Scheeres, 2004a), this effect does not necessarily lead to the excess of slow rotators in the NEA population, as noted in Harris (2002). It should be noted that YORP, by itself, should be able to produce such extremely slow rotation in asteroids.

Although escape of the secondary may occur, it does not necessarily have to occur. Another possibility is that the secondary reimpacts on the primary. If nothing changes with these two bodies, however, there is too much angular momentum in the system for them to lie on each other, and they would enter orbit again. Alternatively, it is plausible that the bodies may lose some of their loose components or separate into smaller pieces. This will essentially create a system with a smaller mass fraction and, potentially, a higher fission rate, as inferred from Fig. 6. A system following this pathway may redistribute its

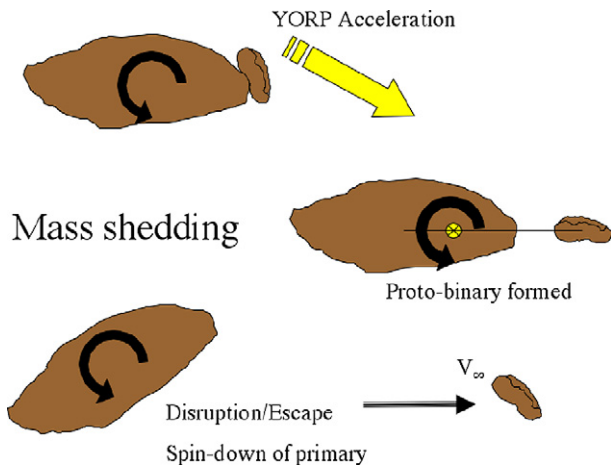


Fig. 15. Spin regulation cycle. A parent contact binary is spun to fission. If the primary has an elongate shape and the spun off particle is small, the resulting relative equilibrium will be unstable and may have positive energy. If so, in the resulting evolution the fissioned secondary may escape and decrease the rotation rate of the primary (Harris, 2002).

mass via successive breakage of its components until it essentially resembles an ellipsoid.

5.2. Contact binary cycle

As we consider bodies with a larger mass fraction we move into a realm where the system may no longer have a positive free energy, but where the resultant relative equilibrium that fissions is still unstable. It is instructive to approximate the Asteroid Itokawa as being comprised of an ellipsoidal body and a spherical head. In Demura et al. (2006) the relative dimensions of these portions of the body are given: body $490 \times 310 \times 260$ m, head $230 \times 200 \times 180$. As the body is much more elongate than the head, we make the approximation that the body is an ellipsoid and the head a sphere. If we formally compute the fission limit for this system and evaluate its stability we find that it is an unstable relative equilibrium which lies just within the negative energy curve. Thus if the asteroid experiences fission it would be impossible for it to escape, yet the resultant relative equilibrium would also be very unstable and could lead to reimpact.

In this contact binary cycle, first proposed in Scheeres et al. (2007), the body is accelerated until it achieves an energetic transition. Such a transition would involve the competent portions of the body moving relative to each other, most likely stirring up considerable dust and ejecta and resurfacing and shaking the entire system. Such configuration changes may easily change the net effect of the YORP torque, and place the body into a deceleration mode which would decrease its angular momentum for a time. Should this not occur, and the body continue to gain angular momentum, eventually it would fission into an unstable, but bounded orbit. A likely outcome in this state is a re-impact of the portions or potentially ejection of smaller mass that was lofted into orbit due to this shifting. Resulting from this could either be a sufficient loss of angular momentum to allow re-impact, or a re-impact with a negative net YORP torque, causing the system to decelerate. The theory

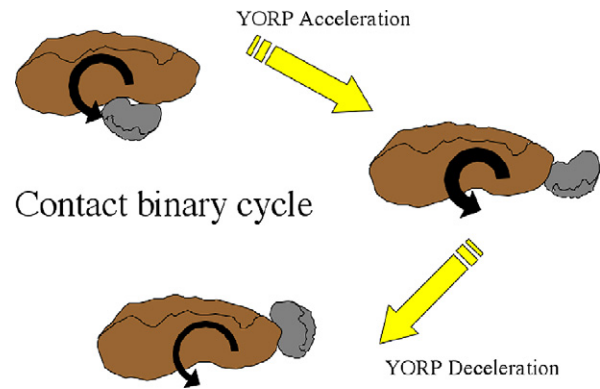


Fig. 16. Contact binary cycle. A parent contact binary is spun to fission. If the resulting relative equilibrium is unstable but the free energy is negative, the system cannot disrupt. If so, in the resulting evolution the secondary may reimpact on the primary and reform a contact binary asteroid (Scheeres et al., 2007).

of YORP acceleration predicts that Itokawa should have gone through such a cycle in the recent past, which is entirely consistent with its surface state. Specifically, consistent with the finest regolith lying at the lowest points of the body, consistent with its boulder-strewn surface, and consistent with the apparent presence of young and old boulders in close proximity to each other (Fujiwara et al., 2006). The time-scale for Itokawa to have been in this state is only on the order of a few hundred thousand years, according to computations (Scheeres et al., 2007).

In this scenario, depicted in Fig. 16, we note that contact binary asteroids with the “wrong” primary shape and mass fraction can be trapped in a repeating cycle of spin-up to fission or near fission, followed by an eventual spin-down once the system has reconfigured itself to have a net YORP deceleration configuration. While a system may be trapped in such a cycle for a time period, there are ways in which it could escape from it. First, during every fission event it is likely that smaller ejecta may be liberated and may transition to escape, thus slowly changing the mass distribution and characteristics of the asteroid system. Second, if the system experiences a rapid period of energy dissipation after fission, it may be possible for the system to relax into a stable relative equilibrium state. In this context we note the existence of the exterior unstable manifolds for the unstable system with negative energy (Fig. 10). A contact binary system that falls into this mode may be able to dissipate energy rapidly enough to completely stabilize the system. In the opposite direction, if the secondary is fractured upon impact into smaller portions, the system can evolve towards the spin-regulation cycle. These are all questions of interest that require further study.

5.3. Stable binary creation

Finally we move to larger values of ν and see that fission will occur in the stable regions of Figs. 7 and 8. This leads to the birth of stable binary asteroids from fission (see Fig. 17). We note that the characteristics of most observed NEO binaries (Pravec et al., 2006) fit with the expectations from the theory and include nearly spherical primaries and rapid rotation. It is

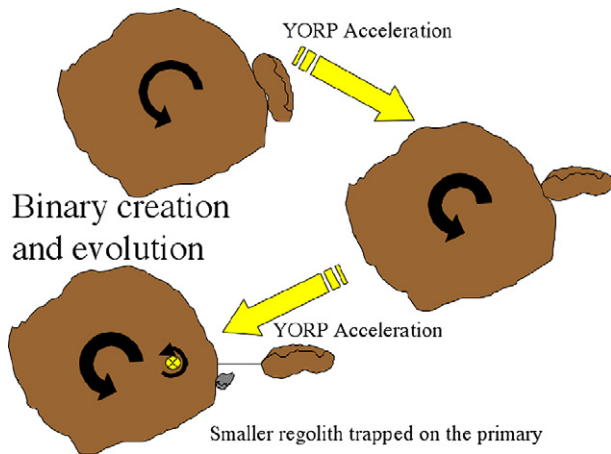


Fig. 17. Binary birth. A parent contact binary is spun to fission. If the resulting relative equilibrium is stable, the system is configuration stable. Also, the system's total free energy is negative and cannot disrupt. Thus, once formed the system can persist over an extended period of time, during which other evolutionary mechanisms become active (Scheeres et al., 2006).

important to note that there are no observed stable binaries in the NEO population that fall into the other range of the stability diagram. The mechanism here is rather simple. The system is accelerated until it achieves fission in the minimum energy (1, 3) configuration. For a small enough elongate body (ν large enough) this system is energetically stable and should survive.

After the birth of a stable binary system, it will continue to evolve due to several phenomena. These include tidal dissipation, unless the system is completely relaxed, additional torques and forces due to solar radiation pressure, and occasional planetary flybys. The subsequent pathways it may take are an item for future study, and are outlined in Scheeres (2007). There are apparently three main evolutionary paths. First, external perturbations may drain angular momentum away from the system, and may cause the two bodies to reinitiate contact and fall back into a contact binary body (Ćuk and Burns, 2005). Second, we note that the bodies are born into a mutually synchronous configuration, and that this mutual synchronous configuration may persist in time. This would mimic the NEA binary Hermes (Margot, 2006), which has nearly equal sized components and is mutually synchronous. Third, if either the orbit is evolved more rapidly than the primary spin state evolution, or vice versa, the primary and relative orbit may fall out of their synchronous lock. Then, the system may mimic the majority of known NEA binaries, with a secondary in synchronous lock and a primary rotating much more rapidly. For this latter case, once the system moves out of its mutually synchronous lock, tidal evolution can act more efficiently, as there will be time-varying body accelerations acting on the system (Scheeres et al., 2006).

From a phenomenological point of view, the latter course appears to be the most common one, and may be tied to the existence of a larger disparity in the mass distribution between the two bodies. If the driving evolution for fission is the YORP effect, then one should expect that binaries with larger mass disparities will be subject to unequal torques, and hence may more easily fall out of a mutually synchronous state. Similarly, so-

lar radiation pressure forces acting on the system may induce orbital evolution rapid enough to break the mutually synchronous lock (Ćuk and Burns, 2005). Conversely, nearly equal sized components may experience similar forces and torques and hence may tend to stay locked in a mutually synchronous situation, as is the case for Hermes.

5.4. Real-world considerations

For real asteroids it is implausible to assume that bodies will neatly slide or roll across each other into new minimum energy configurations once certain energy thresholds are crossed. Rather, surface topography and the rubble pile structure of these bodies themselves will hinder the system from smoothly seeking out its minimum energy state. Also, local concentrations of mass and deviations from such ideal shapes can change the minimum energy configurations in ways that have not been studied to date. Due to these considerations, under an increasing angular momentum load an asteroid may easily be pushed beyond its energetic threshold where it will be lying in a formally unstable state. The system would then be lying in a state similar to a “perched rock,” waiting for a sufficient energy pulse to allow it to seek out its global minimum configuration. Such a transition could be initiated by a small impact or even a relatively distant planetary flyby. If the angular momentum is deposited by a close planetary flyby, it may also coincide with a large scale jostling of the system which may precipitate a change of configuration into its minimum energy state at the same time that the system is given a new minimum energy state.

For either scenario, gradual or abrupt change, global transitions in the system configuration can occur with components being transferred from one region to another. If these components are rubble piles, then as they move relative to each other they should also lose material from their surface leaving boulders and other debris from one component on the other. This would provide a surface mixture of rubble from both bodies. It will also expose previously covered material which would have been, by definition, in the interior and cover material that previously was on the surface. Furthermore, the actual shapes of the components may change during these transition periods.

Such reconfigurations can work in either direction; should the angular momentum of an asteroid be decreased due to YORP or a flyby the system can also collapse back into a low-rotation rate minimum energy configuration. Thus over long time spans it is possible for an asteroid consisting of multiple components to have them shift back and forth in a cycle of changing minimum energy configurations. This action would tend to mix material from the interior and surface regions, implying that when we look at an asteroid we look at its insides as well as its outside, with implications for sample return missions to small NEA.

The current paper presents an idealized model for the evolution of asteroids composed of multiple pieces. While simple, the basic mechanics of these systems are already complex for the models considered here. To extend these analyses beyond simple ellipsoid–sphere systems will require numerical analysis and more sophisticated physical models.

6. Conclusions

A general description of the minimum energy configurations for a contact binary has been given as a function of its total angular momentum. If the total angular momentum of a contact binary system evolves in time due to external perturbations, we find that the minimum energy configuration of the system can have drastic shifts. Given external excitation, it is feasible for a contact binary asteroid to transition from one configuration to another, in the process mixing and disturbing the surface of the body and creating a great deal of “seismic shaking.” If the angular momentum of the contact binary increases enough, we find that the minimum energy configuration for almost all systems has the elongate body’s long axis pointed towards the spherical body’s center. If the angular momentum continues to increase this configuration can naturally evolve into an orbital relative equilibrium.

Such an orbital equilibrium is not always stable, however, which leads to three different types of evolution that can occur for a contact binary spun to fission. If the primary is elongate and dominates the mass of the system, the resulting orbital system is highly unstable and the two components can escape from each other. The net result of this is that the spin rate of the primary will be decreased, and its largest component will no longer be bound to the body. Conversely, if the secondary reimpacts the primary and breaks into smaller pieces, the spin rate can continue to increase until the next largest component reaches fission rates. Due to the much lower rotation rates that a contact binary will fission at, as compared to the classical model of a pebble on the surface of a spinning sphere, this result implies that NEO below the classical rubble-pile spin rate barrier may actually be at or close to a fission limit for a contact binary body.

If the primary is still elongate but the secondary has a larger share of the mass (on the order of 20–40% of the total mass), the orbit that results from fission is still unstable but now the system does not have sufficient energy for the two bodies to escape from each other. The system can then re-impact and fracture, changing the mass distribution of the system. Such a reconfiguration may change the effect of YORP and cause the net rotational accelerations to become negative and decrease the system’s angular momentum. Otherwise, if sufficient energy can be dissipated in a short time span the system may become trapped into a stable relative equilibrium. Except for the final option, the other two force the system to remain a contact binary.

Finally, if the primary is spheroidal and contains most of the mass of the system, how much depending on the shape of the ellipsoid, the fissioned system will directly transition into a stable binary system. Depending on the net torques acting on each body and the forces acting on the relative orbit, the two components may continue their evolution and remain synchronously locked (if their masses are close to each other in size) or may fall out of mutual synchronous lock, with the smaller elongate body remaining in a synchronous orbit.

These different evolutionary paths are speculative, but are based on the current model used in this paper. It is significant

to note that the qualitative features of the NEO asteroid population, including binaries, is consistent with this model. Future work must generalize the model to better mimic actual asteroid shapes, test some of the evolutionary models with numerical integrations, and further develop how shape transitions could occur once the system is placed into a non-minimum energy state.

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